SLEEPING MACHINES • TUTORIAL 6

Counting Processes and Fisher Geometry

Compensators, timed races, and natural gradients with explicit parameter boundaries

Why this tutorial

Race learning observes both *which* clock fired and *when*. The waiting time supplies information that a winner-only classifier discards. Counting-process martingales reveal this information geometry and explain when a diagonal credit preconditioner is exact.

1 Counts, compensators and martingales

Let $N_i(t)$ count observed events of type i, with predictable intensity $\lambda_i(t \mid H_{t^-})$. Its compensator is $A_i(t) = \int_0^t \lambda_i(u)\,\mathrm{d}u$. Under the usual integrability assumptions,

$$M_i(t) = N_i(t) - A_i(t)$$

is a martingale: conditional expected future increments are zero. "Predictable" means the intensity is determined by the history before the event; using the arriving mark to define its own prior intensity would violate causality.

Quadratic variation. Unit jumps give $[M_i]_t = N_i(t)$ and predictable variation $\langle M_i\rangle_t = A_i(t)$. If two type processes have no simultaneous jumps, $[M_i, M_j]_t = 0$ for $i \neq j$, and their compensated martingales are orthogonal. Physical timestamp ties in recorded data are not automatically independent simultaneous continuous-time jumps: use a declared event order or an appropriate joint observation model.

2 Scores as martingale integrals

Consider fixed observation horizon T and log-scale coordinates

$$\lambda_i(t;\theta) = e^{\theta_i} b_i(t, H_{t^-}),$$

where each predictable b_i is a fixed function of observed history and does not otherwise depend on any θ_j. The log-likelihood is

$$\ell(\theta) = \sum_i \int_0^T \log \lambda_i(t;\theta)\,\mathrm{d}N_i(t) - \sum_i \int_0^T \lambda_i(t;\theta)\,\mathrm{d}t.$$

Thus $U_i = \partial_{\theta_i}\ell = N_i(T) - A_i(T) = M_i(T)$. Martingale isometry and orthogonality give

$$F_{ij} = \mathbb{E}[U_i U_j] = \delta_{ij}\mathbb{E}[A_i(T)] = \delta_{ij}\mathbb{E}[N_i(T)].$$

The same identity holds for a stopped first-event experiment under the required optional-stopping and integrability conditions. Censored no-event observations contribute $-A_i$ to the score: silence is part of the information.

Key idea

Diagonal Fisher information is exact for these separate log-intensity scale coordinates and observed event causes. It is not diagonal for every weight in a deep race network, or for an unobserved latent winner after marginalization.

3 A worked exponential race

For independent exponential clocks with rates $\lambda_i = e^{\theta_i}$, observe the first time τ and winner I. Write $\Lambda = \sum_i \lambda_i$ and $p_i = \lambda_i/\Lambda$. Then

$$\ell = \theta_I - \Lambda\tau, \qquad U_i = \mathbf{1}_{I=i} - \lambda_i\tau.$$

Winner and time are independent, $\mathbb{E}\tau = 1/\Lambda$, and $\mathbb{E}\tau^2 = 2/\Lambda^2$. For $i \neq j$,

$$\mathbb{E}[U_iU_j] = -\lambda_j\frac{p_i}{\Lambda} - \lambda_i\frac{p_j}{\Lambda} + \lambda_i\lambda_j\frac{2}{\Lambda^2} = 0.$$

On the diagonal, $p_i - 2\lambda_i p_i/\Lambda + 2\lambda_i^2/\Lambda^2 = p_i$. Hence $F = \mathrm{diag}(p)$.

The softmax gauge. If only the winner is observed, $F_{\text{winner}} = \mathrm{diag}(p) - pp^\top$. Adding a common constant to all θ_i leaves winner probabilities unchanged, so the all-ones direction is null. The timed race also supplies $pp^\top$, recovering $F = \mathrm{diag}(p)$. Time identifies the total rate.

4 Natural gradients and parameter sharing

Natural-gradient ascent uses $\Delta\theta = \eta F^{-1}U$. For independent race log-scales, this normalizes each coordinate by its expected win exposure. In practice, estimate exposure from past data and regularize near-zero entries; rare clocks otherwise produce enormous updates.

Cramér–Rao. For a locally unbiased estimator from n independent races with identifiable parameters, $\mathrm{Cov}(\widehat{\theta}) \succeq F^{-1}/n$. A rarely winning clock has low information per observation. Stored capacity can therefore exceed statistically supported capacity even when dormant clocks are cheap to retain.

Deep parameters. If log-scales depend on shared weights w, with Jacobian $J = \partial\theta/\partial w$, the conditional pullback is $J^\top FJ$, generally dense. For history-dependent features, sum the corresponding event/compensator contributions with their own Jacobians. Generalized Gauss–Newton uses analogous output-space curvature pullbacks; the exact Hessian additionally contains second derivatives of the parameter map. Learning memory transport or hidden routes requires their credit paths too.

Exercises

Exercise 1. A two-clock race has rates $(2, 1)$. Compute the timed and winner-only Fishers.

Solution. Timed: $\mathrm{diag}(2/3, 1/3)$. Winner-only: $\left(\begin{smallmatrix} 2/9 & -2/9 \\ -2/9 & 2/9 \end{smallmatrix}\right)$, whose row sums vanish.

Exercise 2. No event occurs before censoring time T in an exponential race. What is the score?

Solution. $\ell = -\Lambda T$ and $U_i = -\lambda_i T$. Omitting this silence term biases learning toward excessive event rates.

Exercise 3. Both clock scales share one weight: $\theta_1 = w$, $\theta_2 = 2w$. Find the scalar Fisher.

Solution. $J = (1, 2)^\top$, so $F_w = p_1 + 4p_2$. The diagonal race-coordinate result has been pulled back through a shared parameterization.

Where this appears in our work

Note 168 and credit/race_fisher_mc.py study timed-race geometry. RaceFisher.lean checks the exponential-race algebra given the stated moments; it does not formalize compensators or optional stopping. Native clock/mark heads share temporal and addressed value features. A diagonal local update must therefore be labelled a local coordinate preconditioner unless the complete pullback is accounted for.

Further reading. Note 168, *Race information geometry*, and its witness and Lean algebra; tutorial 1 for race survival and joint densities.